

Categorical Time-Series Analysis

General Notation

(X_t) is a categorical (ordinal/ nominal) *stationary* process

1. $E(X_t) = \mu_X \quad \forall t$
2. $E(X_t^2) < \infty \quad \forall t$
3. $Cov(X_t, X_{t+h}) = \gamma(h) \quad \forall t, h$

with range $S = \{s_0, \dots, s_m\}$ $m \in \mathbb{N}$. While the order of the elements in S represents the natural of the elements in the ordinal case, it is arbitrary in the case of nominal S and only used for readability and consistency in notation.

Analytical tools for ordinal time series are using the marginal CDF ($\mathbf{f}$) while tools for nominal data are based on the marginal PMF ($\mathbf{p}$):

$$\begin{aligned} \mathbf{f} &:= F(X_{t \text{ (ord.)}}) = (f_0, \dots, f_{m-1})^T = (P(X_{t \text{ (ord.)}} \leq s_0), \dots, P(X_{t \text{ (ord.)}} \leq s_{m-1}))^T \\ \mathbf{p} &:= f(X_{t \text{ (nom.)}}) = (p_0, \dots, p_m)^T = (P(X_{t \text{ (nom.)}} = s_0), \dots, P(X_{t \text{ (nom.)}} = s_m))^T \end{aligned}$$

And bivariate (cumulative) probabilities with time lag (h) are accordingly:

$$\begin{aligned} f_{ij}(h) &= P(X_t \leq s_i, X_{t+h} \leq s_j) \quad j, i < m \\ p_{ij}(h) &= P(X_t = s_i, X_{t+h} = s_j) \quad j, i \leq m \end{aligned}$$

Location

For ordinal data, the measure of location is given by the median and for ordinal its the mode

$$\begin{aligned} Med(X_{t \text{ (ord.)}}) &= \min(s_i \in S : F(s_i) \geq \frac{1}{2}) \\ Mod(X_{t \text{ (nom.)}}) &= s_i \in S : P(X_t = s_i) = \operatorname{argmax}(\mathbf{p}) \end{aligned}$$

Measures of Dispersion

Dispersion of categorical variables is commonly quantified via the *Index of Ordinal Variation* (IOV) for ordinal data and the *Index of Qualitative Variation* (IQV) for nominal data. Both measures take values in the interval $[0, 1]$, where 0 indicates no dispersion (degenerate one-point distribution) and 1 indicates maximal dispersion.

For ordinal variables, dispersion is defined in terms of the CDF vector $\mathbf{f}$:

$$\text{IOV} = \frac{4}{m} \sum_{i=0}^{m-1} f_i(1 - f_i),$$

where $f_i = P(X_t \leq s_i)$.

For nominal variables, dispersion is defined in terms of the PMF $\mathbf{p}$:

$$\text{IQV} = \frac{m+1}{m} \left(1 - \sum_{i=0}^m p_i^2 \right), \quad p_i = P(X_t = s_i).$$

Minimal Dispersion

Both measures attain the value 0 if and only if the distribution is degenerate, that is, if all probability mass is concentrated in a single category.

Ordinal case. A one-point distribution is characterized by a CDF vector $\mathbf{f}$ of the form

$$\mathbf{f} \in \left\{ \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\}.$$

In each case, at every index i either $f_i = 0$ or $f_i = 1$, hence

$$f_i(1 - f_i) = 0 \quad \forall i.$$

Therefore,

$$\text{IOV} = \frac{4}{m} \sum_{i=0}^{m-1} f_i(1 - f_i) = 0.$$

Nominal case. A degenerate distribution has a PMF of the form

$$\mathbf{p} \in \left\{ \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\}.$$

Thus,

$$\sum_{i=0}^m p_i^2 = m \cdot 0 + 1 \cdot 1 = 1,$$

and therefore

$$\text{IQV} = \frac{m+1}{m} (1 - 1) = 0.$$

Maximal Dispersion

Ordinal case. Dispersion of an ordinal variable is maximal when half of the probability mass lies in the lowest category and half in the highest category; all intermediate categories receive probability zero. This results in a CDF vector

$$\mathbf{f} = \left(\frac{1}{2}, \frac{1}{2}, \dots, \frac{1}{2} \right)^\top.$$

Then

$$f_i(1 - f_i) = \frac{1}{2} \left(1 - \frac{1}{2}\right) = \frac{1}{4} \quad \forall i,$$

and thus

$$\text{IOV} = \frac{4}{m} \cdot \sum_{i=0}^{m-1} \frac{1}{4} = \frac{4}{m} \cdot \frac{m}{4} = 1.$$

Nominal case. Maximal dispersion is obtained when probability mass is distributed uniformly across all $m + 1$ categories:

$$p_i = \frac{1}{m + 1}, \quad i = 0, \dots, m.$$

Then

$$\sum_{i=0}^m p_i^2 = (m + 1) \frac{1}{(m + 1)^2} = \frac{1}{m + 1},$$

so that

$$\text{IQV} = \frac{m + 1}{m} \left(1 - \frac{1}{m + 1}\right) = \frac{m + 1}{m} \cdot \frac{m}{m + 1} = 1.$$

Samples Measures

For the estimation of the (cumulative) probabilities based on samples we use the (cumulative) relative frequencies, that can be best expressed if switching from (X_t) to its binarization ($(\mathbf{Z}_t)$ for ordinal variables, $(\mathbf{Y}_t)$ for nominal):

$$\mathbf{Z}_t = (Z_{t,0}, \dots, Z_{t,m-1})^T, Z_{t,i} = I_{\{X_t \leq s_i\}} \quad \mathbf{Y}_t = (Y_{t,0}, \dots, Y_{t,m})^T, Y_{t,i} = I_{\{X_t = s_i\}}$$

Now we can estimate $\mathbf{p}/\mathbf{f}$ by taking the mean about the binarization:

$$\hat{\mathbf{f}} = \frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t, \quad \hat{\mathbf{p}} = \frac{1}{n} \sum_{t=1}^n \mathbf{Y}_t$$